Note on: *Everything You Always Wanted to Know About Multiple Interest Rate Curve Bootstrapping But Were Afraid To Ask*

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Reading notes

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One-paragraph summary

A comprehensive practitioner-grade treatment of post-crisis interest rate yield curve construction: how to build a discount curve and one or more FRA curves consistent with the diffusion of collateral agreements (CSA) and the persistence of large Basis-Swap spreads between tenors. The theoretical core (§3): under a perfect CSA with collateral rate r_c, the value of a derivative on a generic underlying X obeys the PDE $\partial_t \Pi + r_f X \partial_X \Pi + \frac{1}{2}\sigma^2 X^2 \partial^2_{XX} \Pi = r_c \Pi$ — *drift uses the funding rate r_f, discount uses the collateral rate r_c* (1). The OIS curve plays the role of discounting curve; tenor-specific FRA curves play the role of forwarding curves. The practical core (§4): the bootstrap is a recursive application of pricing formulas to a curated set of market instruments (Deposits, FRAs, Futures, IRS, OIS, Basis Swaps), with attention to interpolation, synthetic instruments for short-end coverage, negative rates (post-2012 EUR), turn-of-year effect, and the loss of the classical telescoping identity for IRS rates (9). The implementation core (§5): a full EUR-market case study (11-Dec-2012) bootstrapping the Eonia discount curve plus four FRA curves (1M, 3M, 6M, 12M), with all code released open-source in QuantLib.

1 Setup and notation

Paper	House	Meaning
$B_\alpha(t)$	—	Generic funding account at rate r_α
$B_f(t)$	—	Treasury (unsecured) funding account, rate r_f
$B_c(t)$	$B_c(t)$	Collateral account, rate r_c (typically Eonia / OIS)
$r_f(t)$	r^{fun}	Unsecured funding rate (Libor + spread)
$r_c(t)$	r^{c}	Collateral rate (OIS, set by CSA)
$\Pi(t,X)$	—	Derivative price on underlying X
$P_c(t,T)$	$P_c(t,T)$	Collateral zero-coupon bond, discount factor on $\mathcal{C}_c$
$P_x(t,T)$	$P_x(t,T)$	Forward-curve zero-coupon factor (tenor x)
$L_x(T_{i-1},T_i)$	—	Spot Libor rate for tenor x
$F_x(t;T_{i-1},T_i) = F_{x,i}(t)$	—	Forward Libor rate (tenor x) at t
Q_f	—	Probability measure with numeraire B_f
Q_f^T	—	T-forward measure under $P_c(t,T)$
$\tau(T_a,T_b,dc)$	α_i	Year fraction
$\mathcal{C}_x$	$\mathcal{C}_x$	Yield curve indexed by tenor x (e.g. $\mathcal{C}_{6M}^{EUR}$)
$\mathcal{C}_c$	$\mathcal{C}_c$	Discounting (OIS / collateral) yield curve
$\mathcal{C}_x^P, \mathcal{C}_x^F$	—	Discount-factor curve, FRA curve representations of $\mathcal{C}_x$
$A_c(t;\mathcal{S})$	A_c	Collateral annuity $\sum_j P_c(t,S_j)\tau_K(S_{j-1},S_j)$
$C_{c,x}^{\text{FRA}}(t;T)$	—	Convexity adjustment between $\mathcal{C}_c$ and $\mathcal{C}_x$
$R_{\text{on}}^{\text{OIS}}(t;\mathcal{T},\mathcal{S})$	—	Par OIS rate
$R_x^{\text{IRS}}(t;\mathcal{T},\mathcal{S})$	—	Par IRS rate (tenor x floating leg)
$\Delta(t;\mathcal{T}_x,\mathcal{T}_y,\mathcal{S})$	—	Par Basis-Swap spread (tenor x vs. y)
ω	—	Payer/receiver indicator (± 1)

2 Key equations

Pricing under perfect CSA (§3.3, eq. 9):

$$\frac{\partial \Pi}{\partial t} + r_f(t)\,X(t)\,\frac{\partial \Pi}{\partial X} + \tfrac{1}{2}\,\sigma^2(t,X)\,X^2(t)\,\frac{\partial^2 \Pi}{\partial X^2} = r_c(t)\,\Pi(t,X). \tag{1}$$

Says: the drift on the underlying uses the unsecured funding rate r_f (the rate at which the hedge is financed), while the discount rate on the derivative is the collateral rate r_c (the rate the cash collateral earns). The two rates are now genuinely different and not interchangeable. Feynman-Kac gives

$$\Pi(t,X) = \mathbb{E}_t^{Q_f}\left[\exp\left(-\int_t^T r_c(u)\,\mathrm{d}u\right)\Pi(T,X)\right], \tag{2}$$

under the funding measure Q_f where X has drift $r_f X$. The change of numeraire from B_c to $P_c(t,T)$ gives the cleaner forward-measure expression:

$$\Pi(t,X) = P_c(t,T)\,\mathbb{E}_t^{Q_f^T}[\Pi(T,X)], \qquad P_c(t,T) = \mathbb{E}_t^{Q_f}\left[\exp\left(-\int_t^T r_c(u)\,\mathrm{d}u\right)\right]. \tag{3}$$

Says: discounting is done at the collateral rate, hence the “OIS discounting” name for the post-crisis convention.

Deposit pricing and bootstrap relation (§4.3.1, eq. 40–41):

$$\mathrm{Depo}(t;T_i) \;=\; N\,P_x(t,T_i)\,[1+R_x^{\mathrm{Depo}}(T_0^F;T_i)\,\tau_L(T_0,T_i)], \tag{4}$$

$$P_x(T_0,T_i) \;=\; \frac{1}{1+R_x^{\mathrm{Depo}}(t_0;T_i)\,\tau_L(T_0,T_i)}. \tag{5}$$

Says: Deposits are unsecured, so the discount factor used is the tenor-x curve factor P_x (not the collateral factor P_c). Eq. 5 bootstraps a short-end pillar of $\mathcal{C}_x$ from the quoted deposit rate.

Market FRA pricing (§4.3.2, eq. 43–44):

$$\mathrm{FRA}^{\mathrm{Mkt}}(t;\mathcal{T},K,\omega) \;=\; N\omega\,P_c(t,T_{i-1})\left[1 \;-\; \frac{1+K\,\tau_x(T_{i-1},T_i)}{1+F_{x,i}(t)\,\tau_x(T_{i-1},T_i)}\,e^{C_{c,x}^{\mathrm{FRA}}(t;T_{i-1})}\right]. \tag{6}$$

Says: the actual market-FRA payoff is paid at T_{i-1} (start of accrual), not at T_i as in textbook FRAs — so the exposure-resetting convention introduces a forward-discount-factor ratio. The convexity adjustment $C_{c,x}^{\mathrm{FRA}}$ accounts for the correlation between $F_{d,i}$ (the discount-curve forward) and $F_{x,i}$ (the tenor-x forward); small in typical post-crisis markets.

IRS pricing under multicurve (§4.3.4, eq. 61–63):

$$\mathrm{IRS}(t;\mathcal{T},\mathcal{S},L_x,K,\omega) \;=\; \omega N\,[R_x^{\mathrm{IRS}}(t;\mathcal{T},\mathcal{S})-K]\,A_c(t;\mathcal{S}), \tag{7}$$

$$R_x^{\mathrm{IRS}}(t;\mathcal{T},\mathcal{S}) \;=\; \frac{\sum_{i=1}^{n} P_c(t,T_i)\,F_{x,i}(t)\,\tau_L(T_{i-1},T_i)}{A_c(t;\mathcal{S})}, \qquad A_c(t;\mathcal{S}) \;=\; \sum_{j=1}^{m} P_c(t,S_j)\,\tau_K(S_{j-1},S_j). \tag{8}$$

Says: the par IRS rate is the ratio of the floating leg PV to the fixed-leg annuity. Both legs are discounted on the *collateral* curve P_c, but the floating leg uses the *tenor-x forward* $F_{x,i}$. *Two curves, one rate.*

Loss of the classical telescoping identity (§4.3.4, eq. 64–65):

$$\sum_{i=1}^{n} P_c(t,T_i)\,F_{x,i}(t)\,\tau_L(T_{i-1},T_i) \;\neq\; P_c(t,T_0) \;-\; P_c(t,T_n). \tag{9}$$

Says: in the single-curve world this identity held exactly (because $F_i = (P_{i-1}/P_i - 1)/\alpha_i$); in the multicurve world it fails because the discount curve and the forward curve are different. A spot-starting at-par IRS no longer has each leg worth 1; only the *difference* is zero.

Bootstrap formula for IRS pillar (§4.3.4, eq. 66):

$$F_{x,i}(T_0) \;=\; \frac{R_x^{\mathrm{IRS}}(T_0;T_i)\,A_c(T_0;T_i) \;-\; \sum_{\alpha=1}^{i-1} P_c(T_0,T_\alpha)\,F_{x,\alpha}(T_0)\,\tau_L(T_{\alpha-1},T_\alpha)}{P_c(T_0,T_i)\,\tau_L(T_{i-1},T_i)}. \tag{10}$$

Says: solve for the next forward $F_{x,i}$ given known earlier forwards and an IRS quote. The presence of A_c on the RHS means the bootstrap of $\mathcal{C}_x$ requires the discount curve $\mathcal{C}_c$ as input — they are coupled, but sequentially: build $\mathcal{C}_c$ first from OIS, then build each $\mathcal{C}_x$ using $\mathcal{C}_c$.

OIS pricing identity (§4.3.5, eq. 73–74):

$$R_{\mathrm{on}}^{\mathrm{OIS}}(t;\mathcal{T},\mathcal{S}) \;=\; \frac{\sum_{i=1}^{n} P_c(t,T_i)\,R_{\mathrm{on}}(t;T_i)\,\tau_{\mathrm{on}}(T_{i-1},T_i)}{A_c(t;\mathcal{S})} \;=\; \frac{P_c(t,T_0)-P_c(t,T_n)}{A_c(t;\mathcal{S})}. \tag{11}$$

Says: the second equality holds because the OIS *floating* leg is also discounted on the collateral curve — forward overnight rates can be replicated using P_c, so the classical telescoping identity is recovered. This is the *single-curve property of OIS*: OIS reduces to plain discount-curve arithmetic, which is why OIS is used to build the discount curve.

Bootstrap formula for OIS pillar (§4.3.5, eq. 76):

$$P_c(T_0, T_i) = \frac{[R^{\text{OIS}}_{\text{on}}(T_0, T_{i-1}) - R^{\text{OIS}}_{\text{on}}(T_0, T_i)]\, A_c(T_0, T_i) + P_c(T_0, T_{i-1})}{1 + R^{\text{OIS}}_{\text{on}}(T_0, T_i)\, \tau_K(T_{i-1}, T_i)}. \tag{12}$$

Says: solves for the next collateral discount factor given the next OIS quote and the previous one. This is the recursive bootstrap of $\mathcal{C}_c$.

Basis Swap as portfolio of two IRS (§4.3.6, eq. 81):

$$\Delta(t; \mathcal{T}_x, \mathcal{T}_y, \mathcal{S}) = \frac{\text{IRS}_{x,\text{float}}(t; \mathcal{T}_x, L_x) - \text{IRS}_{y,\text{float}}(t; \mathcal{T}_y, L_y)}{N\, A_c(t; \mathcal{S})}. \tag{13}$$

Says: the par Basis-Swap spread is the difference of the two floating-leg PVs divided by the common fixed-leg annuity — analogous to the par IRS rate, but with two floating legs in the numerator. Used to bootstrap $\mathcal{C}_x$ from $\mathcal{C}_y$ (with $\mathcal{C}_y$ already built) when no direct x-tenor IRS is quoted.

3 Derivations & commentary

Commentary

The big idea: pre-crisis, "one curve fits all" — Libor was treated as risk-free, basis spreads were negligible, single-curve discounting and FRA computation. Post-2007: **(i)** Libor became risky (credit/liquidity premium); **(ii)** basis spreads exploded (3M-6M EUR basis hit ~ 40 bp in late 2008); **(iii)** CSA become near-universal so collateral mattered for pricing. The multicurve framework is the no-arbitrage response: one discount curve (collateral / OIS) and a separate forward curve for each Libor tenor. The paper is the most thorough practitioner write-up of the bootstrap mechanics.

Commentary

The PDE (1) is the foundational result. Drift = funding rate (the cost of holding the hedge), discount = collateral rate (the cost of cash). The two were collapsed pre-crisis when "Libor was both". Post-crisis they are genuinely distinct. The same structure appears in Lou (2018) for TRS, in Piterbarg's funding-cost papers, and in the anonymous discounting note (`anon_discounting_textbooks`) which makes the same point in textbook form.

Commentary

Why OIS for discounting? Three reasons: **(1)** The standard CSA pays interest on cash collateral at the overnight rate (Fed Funds, Eonia, SONIA), so r_c = OIS rate by contract definition. **(2)** OIS has the single-curve property (11): forwarding and discounting collapse onto the same curve, so the bootstrap is just discount-factor algebra. **(3)** OIS is liquid out to long maturities post-crisis, so it provides a full term structure of pillars.

Commentary

Bootstrap order matters. Build $\mathcal{C}_c$ first from OIS (12); then build each tenor-x FRA curve $\mathcal{C}_x$ using IRS, FRA, Futures, and Basis-Swap quotes for that tenor, (10), taking $\mathcal{C}_c$ as input. The dependency is one-way: $\mathcal{C}_c$ does *not* depend on any $\mathcal{C}_x$, but every $\mathcal{C}_x$ depends on $\mathcal{C}_c$. (Iterative joint solution is possible but rarely needed since OIS pillars are abundant.)

Commentary

Loss of telescoping is structurally important. In the single-curve world, the floating leg of an IRS is worth $P_c(t,T_0)-P_c(t,T_n)$ identically because $F_i\alpha_i = P_c(t,T_{i-1})/P_c(t,T_i)-1$. In the multicurve world, $F_{x,i} = (P_x(t,T_{i-1})/P_x(t,T_i)-1)/\tau_L$ involves the forward curve P_x, not P_c — so the cancellation doesn't happen. Practitioner consequence: **(i)** a spot-starting at-par IRS does not have each leg equal to par with notional exchange at maturity; **(ii)** valuation libraries written under the old assumption silently produce wrong P&L; **(iii)** basis sensitivities decompose differently — one delta per curve, not one delta total.

Commentary

Interpolation is part of the bootstrap, not after it. Eq. (10) at year i requires $P_c(T_0,T_\alpha)$ at intermediate dates T_α that may not be on the bootstrap grid (e.g. the IRS fixed leg pays annually but the floating leg pays semi-annually — 9.5Y appears in a 10Y IRS bootstrap). Hence the interpolation choice (linear-in-zero-rates, log-linear in discount factors, cubic spline on instantaneous forwards, etc.) *interacts with the bootstrap*, and inconsistent choices produce kinks or non-monotonic FRA curves. The paper recommends log-cubic spline on discount factors as a default.

Commentary

Negative rates (§4.6). EUR Eonia went negative in 2012, killing libraries built on assumed-positive zero rates. The fix is to interpolate *discount factors* (always positive) rather than zero rates: P_c stays well-defined, $\partial_T P_c$ can become positive (giving negative instantaneous forwards) without crashing the interpolator. Quoting from the paper (p. 44): "the simplest remedy is to interpolate the discount factors directly".

Commentary

Synthetic instruments (§4.4). The short end of the FRA curve is sparse: Deposits cover up to 12M, but for tenors $\geq$ 3M the first FRA might not start until 1M or 3M from spot. The paper constructs *synthetic Deposits* (interpolated from the front end of the OIS curve corrected by the basis) and *synthetic FRAs* (interpolated from the longer-tenor quotes) to fill the gap. These are not market-quoted instruments but consistent shadow instruments that maintain a smooth term structure across pillar choices.

4 Validation

Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.
Quantities the paper defines:

- Collateral-discounted derivative price under perfect CSA via the PDE (1) and the corresponding Feynman-Kac/forward-measure representations (2), (3).
- Bootstrap formulas: Deposit (5), IRS (10), OIS (12).

- Par IRS rate (8), par OIS rate (11), par Basis-Swap spread (13).
- Market-FRA pricing (6) (including the convexity adjustment $C_{c,x}^{\text{FRA}}$).
- Discounting curve $\mathcal{C}_c$ and tenor-specific FRA curves $\mathcal{C}_x$ for $x \in \{1M, 3M, 6M, 12M\}$.

Canonical numerical case (§5, EUR market, 11-Dec-2012):

- Reference date t_0 = 11-Dec-2012, spot date T_0 = 13-Dec-2012.
- Calendar: TARGET. Day count for floating legs: act/360. Day count for fixed legs: 30/360 (bond basis).
- Eonia discount curve $\mathcal{C}_{ON}^{EUR}$: bootstrapped from Eonia ON-1Y Deposits + 1Y–30Y OIS strip + ECB-dated forward OIS (handles MPC meeting jumps in front end).
- Euribor1M curve $\mathcal{C}_{1M}^{EUR}$: bootstrapped from synthetic short Deposits + 1M Deposit + short-term 1M IRS + medium/long IRS-1M (derived from IRS-6M + Basis 1M vs. 6M). Last 4 pillars 35–60Y are flat extrapolation from 30Y.
- Euribor3M, Euribor6M, Euribor12M curves similarly, each on its own instrument selection (FRA, Futures, IRS, Basis).
- Sample short-end OIS quotes (Eonia bid–ask–mid, p. 56): ON 0.040%; 1W 0.070%; 1M 0.074%; 3M 0.047%; 6M 0.018%; 1Y 0.000%. *Note negative quotes from 4M onward.*
- Sample IRS-6M quotes (p. 28): 1Y 0.286%; 5Y 0.762%; 10Y 1.584%; 30Y 2.256%.
- Bootstrap engine: open-source QuantLib implementation accompanying the paper.

Equation ↔ quantity map:

Paper eq.	Symbol	Quantity
(1)	PDE	pricing under perfect CSA
(2)–(3)	$\Pi(t, X)$	FK + forward-measure representations
(4)–(5)	Depo, $P_x(T_0, T_i)$	deposit price and bootstrap
(6)	FRA^{Mkt}	market-FRA price with convexity adj.
(7)–(8)	IRS, R_x^{IRS}	multicurve IRS pricing
(9)	telescoping failure	structural identity *not* valid
(10)	$F_{x,i}(T_0)$	IRS pillar bootstrap (next forward)
(11)	$R_{\text{on}}^{\text{OIS}}$	par OIS rate (telescoping recovered)
(12)	$P_c(T_0, T_i)$	OIS pillar bootstrap (next discount)
(13)	Δ	par Basis-Swap spread

Invariants and identities to verify:

1. *Single-curve limit*: at $r_c \equiv r_f$, the PDE (1) reduces to the Black-Scholes-Merton case; IRS formulas reduce to the classical telescoping identity (9) with equality; multicurve and single-curve agree.
2. *OIS single-curve property*: identity $\sum_i P_c(t, T_i) R_{\text{on}}(t; T_i) \tau_{\text{on}} = P_c(t, T_0) - P_c(t, T_n)$ in (11) must hold exactly under the OIS-discounted forward-rate replication.
3. *Bootstrap exact-fit*: re-pricing each input instrument with the bootstrapped curves recovers its market quote (to the bootstrap tolerance, $\sim 10^{-12}$ for a well-conditioned problem).
4. *Curve nesting*: at any pillar T_i on $\mathcal{C}_c$, $P_c(T_0, T_i) > 0$; at any pillar on $\mathcal{C}_x$, $P_x(T_0, T_i) > 0$. Allowed: $P_x > 1$ or P_c non-monotone in T_i in pathological short-end cases (negative rates), but never zero or negative.
5. *Basis-Swap consistency*: with $\mathcal{C}_y$ already built, bootstrapping $\mathcal{C}_x$ from a Basis Swap (x vs. y) quote via (13) and then re-pricing the basis swap on the resulting curves recovers the quote exactly.

6. *FRA convexity adjustment small*: in typical post-crisis EUR market conditions $|C^{\mathrm{FRA}}_{c,x}(t;T_{i-1})| \lesssim 1$ bp; in validation tests, switching it off and on shouldn't move the bootstrap pillars by more than ~ 0.1 bp.
7. *Floating leg PV $\neq$ Notional at par IRS*: for a canonical 10Y IRS-6M at par, the floating-leg PV computed on the bootstrapped curves should differ from 1 by an amount of order $\sum P_c(t,T_i)F_{x,i}\tau - (1-P_c(t,T_n))$ — exactly the structural deviation (9).
8. *Endogenous vs. exogenous bootstrap* (§4.7): the endogenous variant builds $\mathcal{C}_x$ *and* $\mathcal{C}_c$ jointly; the exogenous variant takes $\mathcal{C}_c$ as already built. Both should give the same $\mathcal{C}_x$ to within numerical tolerance, since the $\mathcal{C}_c$-on-OIS bootstrap is unique.

5 Glossary of symbols (paper's notation)

$\mathcal{C}_c, \mathcal{C}_x$	Discount (collateral / OIS) curve, FRA curve for tenor x
$P_c(t,T), P_x(t,T)$	Discount factors on the respective curves
$L_x(T_{i-1}, T_i)$	Spot Libor / Euribor rate for tenor x
$F_{x,i}(t)$	Forward Libor for (T_{i-1}, T_i) at t, tenor x
$r_f(t), r_c(t)$	Unsecured funding rate / collateral rate (overnight)
Q_f, Q_f^T	Funding-account measure / T-forward measure under P_c
$\tau(T_a, T_b, dc)$	Year fraction with day count dc
$\tau_L, \tau_K, \tau_x, \tau_{\mathrm{on}}$	Year fractions for floating, fixed, FRA, OIS legs
$A_c(t;\mathcal{S})$	Collateral annuity $\sum_j P_c(t,S_j)\tau_K$
$R^{\mathrm{OIS}}_{\mathrm{on}}, R^{\mathrm{IRS}}_x$	Par OIS rate, par IRS rate (tenor x)
$\Delta(t;\mathcal{T}_x,\mathcal{T}_y,\mathcal{S})$	Par Basis Swap spread (tenor x vs y)
$C^{\mathrm{FRA}}_{c,x}(t;T)$	FRA convexity adjustment between $\mathcal{C}_c$ and $\mathcal{C}_x$
ω	± 1 payer/receiver flag
N	Notional
$\mathcal{T} = \{T_0,\ldots,T_n\}$	Floating leg schedule
$\mathcal{S} = \{S_0,\ldots,S_m\}$	Fixed leg schedule
ON, TN, SN	Overnight, Tomorrow-Next, Spot-Next (deposit tenors)
Eonia, Euribor, Libor	EUR overnight, EUR Libor, generic Libor
TARGET	Trans-european Automated Real-time Gross settlement (EUR calendar)
QuantLib	Open-source library implementing the algorithms

TODO / Open questions

- Implement OIS bootstrap of $\mathcal{C}^{EUR}_{ON}$ on the 11-Dec-2012 Eonia strip using (12); verify negative short-end discount factors > 1 for $T < 1$Y (because of negative OIS rates).
- Implement IRS-6M bootstrap of $\mathcal{C}^{EUR}_{6M}$ using (10), taking $\mathcal{C}^{EUR}_{ON}$ as input; re-price all 36 IRS-6M instruments to verify exact fit at every pillar.
- Implement multicurve FRA pricing (6) with and without convexity adjustment; quantify the gap on the EUR-3M FRA strip.
- Implement Basis-Swap based bootstrap: given $\mathcal{C}^{EUR}_{6M}$ and Basis-Swap 1M-vs-6M quotes, derive $\mathcal{C}^{EUR}_{1M}$ via (13); cross-check against the IRS-1M-based bootstrap.
- Cross-check loss of telescoping numerically: for a 10Y at-par IRS-6M with notional 1, compute the floating-leg PV; it should equal $1 - P_c(t, 10Y)$ in the single-curve approximation but differ by a measurable amount under multicurve — quantify.
- Compare interpolation schemes (linear-on-zero-rates, log-linear-on-DF, monotonic cubic spline on instantaneous forwards) on the same instrument set; characterise

the differences at intermediate dates (turn of year is the natural stress point).

- Cross-reference with the anonymous discounting note (`anon_discounting_textbooks`): same PDE structure (drift = repo/funding, discount = collateral), different contexts.
- Implement turn-of-year jump handling (§4.8) as a deterministic offset between Dec. 31 and Jan. 1 on the instantaneous forward curve.
- For each of the 4 FRA curves (1M, 3M, 6M, 12M), produce a smooth instantaneous-forward chart; eyeball test for kinks at IMM/turn-of-year/ECB-meeting dates.